§2 量子力学的形式理论

§2.1 态、态矢，Hilbert 空间

1. 态叠加原理：任何一个量子态是其它几个量子态的叠加，
几个量子态的叠加产生新的量子态.

(1) 量子态 ~ $|\alpha\rangle$ ket

① $|\alpha\rangle + |\beta\rangle = |\gamma\rangle$

② $|\alpha\rangle + |\beta\rangle = |\beta\rangle + |\alpha\rangle$,

$(|\alpha\rangle + |\beta\rangle) + |\gamma\rangle = |\alpha\rangle + (|\beta\rangle + |\gamma\rangle)$

③ $\exists\ |\text{null}\rangle$ s.t. $|\alpha\rangle + |\text{null}\rangle = |\alpha\rangle$,

, $\forall |\alpha\rangle$, $\exists\ |\alpha'\rangle$ s.t. $|\alpha\rangle + |\alpha'\rangle = |\text{null}\rangle$

④ $c|\alpha\rangle = |\alpha\rangle c$, $c|\alpha\rangle$ 与 $|\alpha\rangle$ 相同.

⑤ $(cd)|\alpha\rangle = c(d|\alpha\rangle)$

$(c+d)|\alpha\rangle = c|\alpha\rangle + d|\alpha\rangle$

$c(|\alpha\rangle + |\beta\rangle) = c|\alpha\rangle + c|\beta\rangle$

$1|\alpha\rangle = |\alpha\rangle$

(2) 对偶空间 ~ $\langle\alpha|$ bra

一一对应：$|\alpha\rangle \overset{DC}{\longleftrightarrow} \langle\alpha|$

$c_\alpha|\alpha\rangle + c_\beta|\beta\rangle \overset{DC}{\longleftrightarrow} c_\alpha^*\langle\alpha| + c_\beta^*\langle\beta|$

内积：$\langle\beta|\alpha\rangle = \langle|\beta\rangle \cdot \langle|\alpha\rangle$

① $\langle\beta|\alpha\rangle = \langle\alpha|\beta\rangle^*$

② $\langle\alpha|\alpha\rangle \geq 0$

③ $\langle\gamma|(c_\alpha|\alpha\rangle + c_\beta|\beta\rangle) = c_\alpha\langle\gamma|\alpha\rangle + c_\beta\langle\gamma|\beta\rangle$

模：$\||\alpha\rangle\| = \sqrt{\langle\alpha|\alpha\rangle}$

正交：$\langle\alpha|\beta\rangle = 0$

完备性：线性空间中包含系统中所有的量子态.

完备的线性空间是 Hilbert 空间

§ 2.2 线性算符和可观测量

1、算符：对态矢量的变换

线性：$\hat{A}(|\alpha\rangle+|\beta\rangle) = \hat{A}|\alpha\rangle+\hat{A}|\beta\rangle$ ， $(\hat{A}+\hat{B})+\hat{C} = \hat{A}+(\hat{B}+\hat{C})$

$\hat{A}(c|\alpha\rangle) = c\hat{A}|\alpha\rangle$ $\qquad (c\hat{A})|\alpha\rangle = c(\hat{A}|\alpha\rangle)$

相等：$\hat{A} = \hat{B} \Leftrightarrow \hat{A}|\alpha\rangle = \hat{B}|\alpha\rangle \quad \forall |\alpha\rangle$

厄米共轭：$\hat{A}^\dagger := \quad \hat{A}|\alpha\rangle \overset{DC}{\longleftrightarrow} \langle\alpha|\hat{A}^\dagger$

$(\hat{A}|\alpha\rangle)^\dagger = |\alpha\rangle^\dagger \hat{A}^\dagger = \langle\alpha|\hat{A}^\dagger$ $\qquad \underline{\underline{(|\alpha\rangle^\dagger = \langle\alpha|)}}$

$(c_\alpha|\alpha\rangle + c_\beta|\beta\rangle)^\dagger = c_\alpha^*\langle\alpha| + c_\beta^*\langle\beta|$

$(c_A\hat{A} + c_B\hat{B})^\dagger = c_A^*\hat{A} + c_B^*\hat{B}$

$\hat{A}^\dagger = \hat{A} \rightarrow \hat{A}$是厄米算符

乘法：$(\hat{A}\hat{B})|\alpha\rangle = \hat{A}(\hat{B}|\alpha\rangle)$

$\hat{A}\hat{B} \neq \hat{B}\hat{A}$

$\hat{A}(\hat{B}\hat{C}) = (\hat{A}\hat{B})\hat{C}$

$(\hat{A}\hat{B})^\dagger = \hat{B}^\dagger\hat{A}^\dagger$

证：$\langle\alpha|(\hat{A}\hat{B})^\dagger = [(\hat{A}\hat{B})|\alpha\rangle]^\dagger = [\hat{A}(\hat{B}|\alpha\rangle)]^\dagger \quad (\hat{B}|\alpha\rangle)^\dagger\hat{A}^\dagger = \langle\alpha|\hat{B}^\dagger\hat{A}^\dagger$

对易子：$[\hat{A},\hat{B}] = \hat{A}\hat{B} - \hat{B}\hat{A}$ $\qquad [\hat{A},\hat{B}] = 0 \rightarrow [\hat{A},\hat{B}]$对易，$\hat{A}\hat{B} = \hat{B}\hat{A}$

结合律：$\langle\alpha|(\hat{A}|\beta\rangle) = (\langle\alpha|\hat{A})|\beta\rangle = \langle\alpha|\hat{A}|\beta\rangle$

$\hookrightarrow \langle\alpha|\hat{A}|\beta\rangle = \langle\beta|\hat{A}^\dagger|\alpha\rangle^*$

证：$\langle\alpha|\hat{A}|\beta\rangle = \langle\alpha|(\hat{A}|\beta\rangle) = [(\hat{A}|\beta\rangle)^\dagger|\alpha\rangle]^* = \langle\beta|\hat{A}^\dagger|\alpha\rangle^*$

恒等算符：$\hat{I}|\alpha\rangle = |\alpha\rangle$，$\langle\alpha|\hat{I} = \langle\alpha|$

逆算符：$\hat{A}|\alpha\rangle = |\beta\rangle$，$\hat{A}^{-1}|\beta\rangle = |\alpha\rangle$

$\hookrightarrow \hat{A}^{-1}\hat{A} = \hat{I}$

态矢量的外积：$|\alpha\rangle\langle\beta|$

$(|\alpha\rangle\langle\beta|)^\dagger = |\beta\rangle\langle\alpha|$

证：$(|\alpha\rangle\langle\beta|\gamma\rangle)^\dagger = \langle\beta|\gamma\rangle^*\langle\alpha| = \langle\gamma|\beta\rangle\langle\alpha|$

投影算符：$\hat{P}_\alpha = |\alpha\rangle\langle\alpha|$　$\hat{P}_\alpha|\beta\rangle = |\alpha\rangle\langle\alpha|\beta\rangle$

$\hat{P}^+ = \hat{P}$

证：$(|\alpha\rangle\langle\alpha|)^+ = |\alpha\rangle\langle\alpha|$

$\hat{P}^2 = \hat{P}$

证：$|\alpha\rangle\langle\alpha|\alpha\rangle\langle\alpha| = |\alpha\rangle\langle\alpha|$　（$|\alpha\rangle$ 归一化）

反线性算符：$\hat{A}(c_\alpha|\alpha\rangle + c_\beta|\beta\rangle) = c_\alpha^* \hat{A}|\alpha\rangle + c_\beta^* \hat{A}|\beta\rangle$

2. 厄米算符的本征值，本征矢量.

本征方程：$\hat{A}|\alpha\rangle = \alpha|\alpha\rangle$

定理：厄米算符的本征值必为实数

$\underline{\langle\alpha|\hat{A}|\alpha\rangle} = \alpha\underbrace{\langle\alpha|\alpha\rangle}_{\neq 0}$　实数　→ α 为实数

定理：厄米算符属于不同本征值的本征矢量正交.

证：$\begin{cases} \hat{A}|\alpha\rangle = \alpha|\alpha\rangle \\ \hat{A}|\beta\rangle = \beta|\beta\rangle \end{cases}$　$(\hat{A}|\beta\rangle)^+ = \langle\beta|\hat{A}^+ = \langle\beta|\beta^*$

则 $\langle\beta|\hat{A} = \langle\beta|\beta$

$\langle\beta|\hat{A}|\alpha\rangle - \langle\beta|\hat{A}|\alpha\rangle = 0 = (\beta-\alpha)\langle\beta|\alpha\rangle$

$\alpha \neq \beta \rightarrow \langle\beta|\alpha\rangle = 0$

3. 算符函数

解析函数 $f(z)$ 的展开为：

$f(z) = \sum_n f_n z^n \rightarrow f(\hat{A}) = \sum_n f_n \hat{A}^n$

$f(\hat{A})|\alpha\rangle = f(\alpha)|\alpha\rangle$

谱：算符本征值的集合.

$f(z)$ 良定义：收敛半径 $\geq$ 谱半径

3. 测量和可观测量

假设：$\hat{A}$（系统测量后态）= A $\hat{A}$的本征态，

$\hat{A}$的本征态形成完备集

$\hat{A}$为可观测量，$\hat{A}$的观测值为实数 → $\hat{A}$是厄米算符.

$$|\alpha\rangle = \sum_n \psi_n |n\rangle, \quad \psi_n = \langle n|\alpha\rangle$$

测量后坍缩到 $|n\rangle$ 的概率：$P_n = |\langle n|\alpha\rangle|^2$

完备性条件：$\sum_n |n\rangle\langle n| = 1$

期待值：$\langle\hat{A}\rangle_\alpha = \langle\alpha|\hat{A}|\alpha\rangle$

$$= \sum_{n,n'} \langle\alpha|n\rangle\underline{\langle n|\hat{A}|n'\rangle}\langle n'|\alpha\rangle$$

$\langle n|\hat{A}|n'\rangle = a'\langle n|n'\rangle = a'\delta_{nn'}$

$= a_n\delta_{nn'}$

$$= \sum_n a_n |\langle n|\alpha\rangle|^2$$ ，其中 $P_n = |\langle n|\alpha\rangle|^2$

共同本征态：$|\alpha\rangle$为$\hat{A}$,$\hat{B}$的共同本征态，

$\hat{A}|\alpha\rangle = a|\alpha\rangle$，$\hat{B}|\alpha\rangle = b|\alpha\rangle$

↳ $\hat{A}\hat{B}|\alpha\rangle = b\hat{A}|\alpha\rangle = ba|\alpha\rangle = \hat{B}\hat{A}|\alpha\rangle$

$\hat{A}$，$\hat{B}$对易

定理：若$\hat{A}$,$\hat{B}$对易，则必有共同本征态.

证：$|\alpha\rangle$是$\hat{A}$的本征态，

$\hat{A}(\hat{B}|\alpha\rangle) = \hat{B}(\hat{A}|\alpha\rangle) = a(\hat{B}|\alpha\rangle)$

本征值 a 对应的子空间为$\hat{B}$的不变子空间．记为 W_a

↳ 在 W_a 中可找到 $\dim(W_a)$ 个（共同）本征态

取 $w \in W = \{w_1, \dots w_n\}$

$fw \in W = w' = \{w_1' \cdots w_n'\}$

$$\begin{cases} \hat{A}|\alpha\rangle = a|\alpha\rangle \\ \hat{B}|\alpha\rangle = b|\alpha\rangle \end{cases}$$

∵ $\hat{A}$,$\hat{B}$，Herm——

$\langle\alpha|\hat{B} = \langle\alpha|b$，$\langle\alpha|\hat{A} = \langle\alpha|a$

$$\langle\alpha|\hat{A}\hat{B}|\alpha\rangle = ab\langle\alpha|\alpha\rangle$$

定理：$[\hat{F},\hat{A}]=0$，$[\hat{F},\hat{B}]=0$，$[\hat{A},\hat{B}]\neq 0$

则 $\hat{F}$ 的本征值有简并

证：(1) 充分性：

$\hat{F}|\psi\rangle = f|\psi\rangle$，则 $\hat{A}|\psi\rangle$、$\hat{B}|\psi\rangle$ 也是 $\hat{F}$ 本征态.

可选取 ψ 以让 $\hat{A}|\psi\rangle$、$|\psi\rangle$ 是同一个态，此时可使 $\hat{B}|\psi\rangle$ 与 $\hat{A}|\psi\rangle$ 不同.

$\dim W > 1 \to$ 有简并.

(2) 必要：

设 $\hat{F}$ 有两个不简并本征态 $|\psi_1\rangle$，$|\psi_2\rangle$，对应本征值为 f

$$\hat{A} = a_1|\psi_1\rangle\langle\psi_1| + a_2|\psi_2\rangle\langle\psi_2| \quad (a_1\neq a_2)$$

$$\hat{B} = b|\psi_1\rangle\langle\psi_1| + b^*|\psi_2\rangle\langle\psi_2| \quad (b\neq 0)$$

$$\begin{aligned}[\hat{F},\hat{A}] &= \hat{F}\hat{A}-\hat{A}\hat{F}\\ &= \hat{F}(a_1|\psi_1\rangle\langle\psi_1| + a_2|\psi_2\rangle\langle\psi_2|) - (a_1|\psi_1\rangle\langle\psi_1| + a_2|\psi_2\rangle\langle\psi_2|)\hat{F}\\ &= a_1 f|\psi_1\rangle\langle\psi_1| + a_2 f|\psi_2\rangle\langle\psi_2| - a_1 f|\psi_1\rangle\langle\psi_1| + a_2 f|\psi_2\rangle\langle\psi_2|\\ &= 0\end{aligned}$$

$$[\hat{F},\hat{B}] = 0$$

$$\begin{aligned}[\hat{A},\hat{B}] &= a_1 b|\psi_1|^2|\psi_1\rangle\langle\psi_1| + a_2 b^*|\psi_2|^2|\psi_2\rangle\langle\psi_2|\\ &\quad + a_1 b^*\langle\psi_2|\psi_1\rangle|\psi_1\rangle\langle\psi_2| + a_2 b\langle\psi_1|\psi_2\rangle|\psi_2\rangle\langle\psi_1|\\ &\quad - a_1 b|\psi_1|^2|\psi_1\rangle\langle\psi_1| - a_2 b^*|\psi_2|^2|\psi_2\rangle\langle\psi_2|\\ &\quad - b^* a_1\langle\psi_2|\psi_1\rangle|\psi_2\rangle\langle\psi_1| - b a_2\langle\psi_1|\psi_2\rangle|\psi_1\rangle\langle\psi_2|\\ &= (a_1 b^* - a_2 b)\langle\psi_1|\psi_2\rangle(|\psi_1\rangle\langle\psi_2| - |\psi_2\rangle\langle\psi_1|)\\ &\neq 0\end{aligned}$$

4. 不确定关系

$\Delta\hat{A} = \hat{A} - \langle\hat{A}\rangle_\alpha$. 其中 $\langle\hat{A}\rangle_\alpha = \langle\alpha|\hat{A}|\alpha\rangle$

涨落：$\langle(\Delta\hat{A})^2\rangle_\alpha = \langle\hat{A}^2 - 2\hat{A}\langle\hat{A}\rangle_\alpha + \langle\hat{A}\rangle_\alpha^2\rangle_\alpha = \langle\hat{A}^2\rangle_\alpha - \langle\hat{A}\rangle_\alpha^2$

Schwartz 不等式：

$\langle\alpha|\alpha\rangle\langle\beta|\beta\rangle \geq \langle\alpha|\beta\rangle^2$

证：$(\langle\alpha| + c^*\langle\beta|)(|\alpha\rangle + c|\beta\rangle) = |\alpha + c\beta|^2 \geq 0$

令 $c = \dfrac{-\langle\beta|\alpha\rangle}{\langle\beta|\beta\rangle}$

则 $\langle\alpha|\alpha\rangle - \dfrac{(\langle\beta|\alpha\rangle)^2}{\langle\beta|\beta\rangle} \geq 0$, D.

则 $\langle(\Delta\hat{A})^2\rangle_\alpha \langle(\Delta\hat{B})^2\rangle_\alpha \geq |\langle\Delta\hat{A}\Delta\hat{B}\rangle_\alpha|^2$

$\Delta\hat{A}\Delta\hat{B} = \frac{1}{2}[\hat{A},\hat{B}] + \frac{1}{2}\{\hat{A},\hat{B}\}$

其中 $\{\hat{A},\hat{B}\}$ 为反对易子，$\{\hat{A},\hat{B}\} := \hat{A}\hat{B} + \hat{B}\hat{A}$

$*\ [\hat{A},\hat{B}]^\dagger = (\hat{A}\hat{B} - \hat{B}\hat{A})^\dagger = \hat{B}\hat{A} - \hat{A}\hat{B} = -[\hat{A},\hat{B}]$，纯实数

$\{\hat{A},\hat{B}\}^\dagger = (\hat{A}\hat{B} + \hat{B}\hat{A})^\dagger = \hat{B}\hat{A} + \hat{A}\hat{B} = \{\hat{A},\hat{B}\}$，纯虚数

$|\langle\Delta\hat{A}\Delta\hat{B}\rangle_\alpha|^2$

$= |\langle\frac{1}{2}[\Delta\hat{A},\Delta\hat{B}] + \frac{1}{2}\{\Delta\hat{A},\Delta\hat{B}\}\rangle_\alpha|^2$

$= \frac{1}{4}|\langle[\Delta\hat{A},\Delta\hat{B}]\rangle_\alpha|^2 + \frac{1}{4}|\langle\{\Delta\hat{A},\Delta\hat{B}\}\rangle|^2$

则 $\langle(\Delta\hat{A})^2\rangle_\alpha\langle(\Delta\hat{B})^2\rangle_\alpha \geq \frac{1}{4}|\langle[\Delta\hat{A},\Delta\hat{B}]\rangle_\alpha|^2 + \frac{1}{4}|\langle\{\Delta\hat{A},\Delta\hat{B}\}\rangle|^2$

$\geq \frac{1}{4}|\langle[\Delta\hat{A},\Delta\hat{B}]\rangle_\alpha|^2$

§ 2.3 量子化条件

正则量子化：

$[q_\alpha, q_\beta] = 0$, $[p_\alpha, p_\beta] = 0$, $[q_\alpha, p_\beta] = i\hbar\delta_{\alpha\beta}$

↳ $\Delta x \Delta p \geq \frac{1}{2}\hbar$

$[\hat{A}, \hat{B}+\hat{C}] = [\hat{A}, \hat{B}] + [\hat{A}, \hat{C}]$

$$[\hat{A}, \hat{B}\hat{C}] = \hat{A}\hat{B}\hat{C} - \hat{B}\hat{C}\hat{A} = \hat{A}\hat{B}\hat{C} - \hat{B}\hat{A}\hat{C} + \hat{B}\hat{A}\hat{C} - \hat{B}\hat{C}\hat{A}$$

$$= [\hat{A}, \hat{B}]\hat{C} + \hat{B}[\hat{A}, \hat{C}]$$

$\hat{\vec{L}} = \hat{\vec{r}} \times \hat{\vec{p}}$

$\hat{L}_x = \hat{y}\hat{p}_z - \hat{z}\hat{p}_y$, $\hat{L}_y = \hat{z}\hat{p}_x - \hat{x}\hat{p}_z$, $\hat{L}_z = \hat{x}\hat{p}_y - \hat{y}\hat{p}_x$

$[\hat{L}_x, \hat{L}_y] = [\hat{y}\hat{p}_z - \hat{z}\hat{p}_x, \hat{z}\hat{p}_x - \hat{x}\hat{p}_z]$

$\Rightarrow [\hat{L}_\alpha, \hat{L}_\beta] = \epsilon_{\alpha\beta\gamma} i\hbar \hat{L}_\gamma$, $\hat{\vec{L}} \times \hat{\vec{L}} = i\hbar\hat{\vec{L}}$

§ 2.4 表象和表象变换

1. 选择可观测量 $\hat{A}$ 的本征态 $|n\rangle$ 为基矢

$$|\psi\rangle = \sum_n \psi_n |n\rangle = (|1\rangle \cdots |N\rangle) \begin{pmatrix}\psi_1 \\ \vdots \\ \psi_N\end{pmatrix}$$

$\{\psi_n\}$ 称为 $|\psi\rangle$ 在 $\hat{A}$ 表象下的坐标.

表象变换

$$|\psi\rangle = \sum_n \psi_n |n\rangle = \sum_{n'} \psi_n \langle n'|n\rangle |n'\rangle = \sum_{n'} \psi_{n'} |n'\rangle$$

$$(\psi_{n'}) = (\psi_n)\langle n'|n\rangle = U^+(\psi_n)$$

变换矩阵： $U = \langle n|n'\rangle$　　$|n\rangle$, $|n'\rangle$ 分别为变化前、后表象的本征态/向量.

$$|n'\rangle = \sum_n |n\rangle\langle n|n'\rangle$$

则 $\{\psi_{n'}\} = \{\psi_n\}\tilde{U} = \tilde{U}^+\{\psi_n\}$

$\tilde{U}\tilde{U}^+ = 1$，U 为幺正矩阵.

证. $(\tilde{U}\tilde{U}^+)_{ij} = \sum_k U_{ik} U^+_{kj} = \sum_k U_{ik} U^*_{jk}$

$$= \sum_k \langle i|k'\rangle \langle j|k'\rangle^* = \sum_k \langle i|k'\rangle\langle k'|j\rangle$$

$$= \delta_{ij}$$

左表： $\langle\psi| = (\psi_1^* \cdots \psi_N^*) \begin{pmatrix}\langle 1| \\ \\ \langle N|\end{pmatrix}$

内积： $\langle\psi|\phi\rangle = \sum_n \langle\psi|n\rangle\langle n|\phi\rangle$

$$= (\psi_1^* \cdots \psi_N^*) \begin{pmatrix}\phi_1 \\ \vdots \\ \phi_N\end{pmatrix}$$

力学量完全集：

一 上 收 时 对， 力 量 来 示

$\{\hat{A}_1, \hat{A}_2 \cdots\}$

它们的共同本征态可唯一、完备地描述量子态。

2. 算符的表示

$$\hat F|\psi\rangle = |\phi\rangle$$

$$\phi_m = \langle m|\phi\rangle = \langle m|\hat F|\psi\rangle$$

$$= \sum_n \langle m|\hat F|n\rangle\langle n|\psi\rangle$$

$$= \sum_n F_{mn}\psi_n$$

$$\langle m'|\hat F|n'\rangle = \sum_{m,n}\langle m'|m\rangle\langle m|\hat F|n\rangle\langle n|n'\rangle$$

$$= U^*_{mm'}F_{mn}U_{nn'}$$

$$\Leftrightarrow \hat F' = U^\dagger \hat F U$$

若 U 为 $\hat F$ 的本征矢量，则 $\hat F'$ 为 $\hat F$ 经过幺正变换后的对角化矩阵。

例： 2维 Hilbert 空间中有正交归一基矢 $|1\rangle$, $|2\rangle$

算符：$\hat A = \frac{1}{2}(|1\rangle\langle2| + |2\rangle\langle1|)$，$\hat B = \frac{i}{2}(|1\rangle\langle2| - |2\rangle\langle1|)$

(1) 证明：$\hat A$, $\hat B$ 是厄米算符。

(2) 求出 $\hat B$ 表象中 $\hat A$ 的表示

在表象 $|1\rangle$, $|2\rangle$ 下，

$$\hat A = \begin{pmatrix} \langle1|\hat A|1\rangle & \langle1|\hat A|2\rangle \\ \langle2|\hat A|1\rangle & \langle2|\hat A|2\rangle \end{pmatrix} = \frac{1}{2}\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$$

$$\hat B = \begin{pmatrix} \langle1|\hat B|1\rangle & \langle1|\hat B|2\rangle \\ \langle2|\hat B|1\rangle & \langle2|\hat B|2\rangle \end{pmatrix} = \frac{i}{2}\begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$$

求出 $\hat B$ 的本征矢量：

$$|\hat B - \lambda I| = \begin{vmatrix} -\lambda & \frac{i}{2} \\ -\frac{i}{2} & -\lambda \end{vmatrix} = \lambda^2 - \frac{1}{4} = 0 \rightarrow \lambda = \pm\frac{1}{2}$$

$$\begin{pmatrix} -\frac{1}{2} & \frac{i}{2} \\ -\frac{i}{2} & -\frac{1}{2} \end{pmatrix} \rightarrow \begin{pmatrix} -\frac{1}{2} & \frac{i}{2} \\ 0 & 0 \end{pmatrix} \text{①} \times i - \text{②} \rightarrow \frac{1}{\sqrt2}\begin{pmatrix} i \\ 1 \end{pmatrix}$$

$$\begin{pmatrix} \frac{1}{2} & \frac{i}{2} \\ -\frac{i}{2} & \frac{1}{2} \end{pmatrix} \rightarrow \begin{pmatrix} \frac{1}{2} & \frac{i}{2} \\ 0 & 0 \end{pmatrix} \text{①} \times i + \text{②} \rightarrow \frac{1}{\sqrt2}\begin{pmatrix} i \\ -1 \end{pmatrix}$$

$$\hat{A}' = \begin{pmatrix} \langle\lambda_1|\hat{A}|\lambda_1\rangle & \langle\lambda_1|\hat{A}|\lambda_2\rangle \\ \langle\lambda_2|\hat{A}|\lambda_1\rangle & \langle\lambda_2|\hat{A}|\lambda_2\rangle \end{pmatrix} = \begin{pmatrix} i & 0 \\ 0 & -i \end{pmatrix}$$

3. 坐标和动量表象

(1) 坐标算符 $\hat{x}$ 的本征态：$|x\rangle$ ， $\hat{x}|x\rangle = x|x\rangle$

$\hat{x}|x'\rangle$

$x'|x'\rangle$

本征态 $|x'\rangle$ 在表象 $|x\rangle$ 下：$\langle x|x'\rangle = \delta(x-x')$ 正交归一的 $\{|x\rangle\}$ 基底

$\hat{x}$ 的表示：$\langle x|\hat{x}|x'\rangle = x'\langle x|x'\rangle = x\,\delta(x-x')$ ~ $x_i x_j = \delta_{ij}$

完备性条件：$\int dx\, |x\rangle\langle x| = 1$

则 $|\psi\rangle = \int dx\, |x\rangle\langle x|\psi\rangle$,

$\psi(x) = \langle x|\psi\rangle$

(2) 动量算符的表示

引入辅助线性算符 $\hat{\alpha}$ 满足 $\langle \hat{x}|\hat{\alpha}|\psi\rangle = \frac{\partial}{\partial x}\psi(x)$

$$\langle x|\hat{\alpha}\hat{x}|\psi\rangle = \frac{\partial}{\partial x}\langle x|\hat{x}|\psi\rangle = \frac{\partial}{\partial x}\left[x\langle x|\psi\rangle\right]$$

$$= x\frac{\partial}{\partial x}\langle x|\psi\rangle + \langle x|\psi\rangle$$

$$= \langle x|\hat{x}\hat{\alpha}|\psi\rangle + \langle x|\psi\rangle$$

则 $\hat{\alpha}\hat{x} = \hat{x}\hat{\alpha} + 1$, $[\hat{x}, \hat{\alpha}] = -1$ $[\hat{x}, \hat{p}_x] = i\hbar$

$\hat{p}_x = -i\hbar\hat{\alpha}$

$$\langle\phi|\hat{\alpha}|\psi\rangle = \int dx\, \langle\phi|x\rangle\langle x|\hat{\alpha}|\psi\rangle$$

$$= \int dx\, \langle\phi|x\rangle \frac{\partial}{\partial x}\langle x|\psi\rangle$$

$$= \langle\phi|x\rangle\langle x|\psi\rangle\Big|_{-\infty}^{\infty} - \int \langle x|\psi\rangle\frac{\partial}{\partial x}\langle\phi|x\rangle$$

$$= -\int \langle x|\psi\rangle\frac{\partial}{\partial x}\langle\phi|x\rangle$$

$$= -\int dx\langle\psi|x\rangle^*\left(\frac{\partial}{\partial x}\langle x|\phi\rangle\right)^* = -\int dx\, \langle\psi|x\rangle^*\langle x|\alpha|\phi\rangle^*$$

$$= -\int dx\left(\langle\psi|x\rangle\langle x|\hat{\alpha}|\phi\rangle\right)^*$$

$$= -\langle\psi|\hat{\alpha}|\phi\rangle^* = -\langle\phi|\hat{\alpha}^\dagger|\psi\rangle$$

$\Rightarrow$ $\hat{\alpha}^\dagger = -\hat{\alpha}$, $\hat{p}_x^\dagger = \hat{p}_x$

$\hat{p}_x$ 在坐标表象下的表示：

$$\langle x|\hat{p}_x|x'\rangle = -i\hbar\langle x|\hat{\partial}|x'\rangle = -i\hbar\frac{\partial}{\partial x}(x-x')$$

在 $|x\rangle$ 表象下：

$$\langle x|\hat{p}_x|p_x\rangle = \langle x|p_x|p_x\rangle = p_x\langle x|p_x\rangle$$

$|p_x\rangle$ 为 $\hat{p}_x$ 的本征态，

$$= \int \langle x|\hat{p}_x|x'\rangle\langle x'|p_x\rangle\, dx'$$

$$\Rightarrow -i\hbar\int\frac{\partial}{\partial x}\delta(x-x')\langle x'|p_x\rangle\, dx' = p_x\langle x|p_x\rangle$$

$$-i\hbar\frac{\partial}{\partial x}\langle x|p_x\rangle = p_x\langle x|p_x\rangle$$

解得 $p_x(x) = \langle x|p_x\rangle = \frac{1}{\sqrt{2\pi\hbar}}e^{\frac{ip_x x}{\hbar}}$ 傅里叶变换.

动量表象下： $\langle p_x|p_x'\rangle = \delta(p_x-p_x')$

$$\langle p_x|\hat{p}_x|p_x'\rangle = p_x\,\delta(p_x-p_x')$$

又在动量表象的表示：

$$\langle p_x|\hat{x}|p_x'\rangle = \int dx\,dx'\,\langle p_x|x\rangle\langle x|\hat{x}|x'\rangle\langle x'|p_x'\rangle$$

$I = \int dx\,|x\rangle\langle x| = \int dx'\,|x'\rangle\langle x'|$

$$= \int dx\,dx'\,\langle p_x|x\rangle\, x\,\delta(x-x')\,\langle x'|p_x'\rangle$$

$$= \frac{1}{2\pi\hbar}\int dx\,dx'\, x\,\delta(x-x')\, e^{-\frac{ip_x x}{\hbar}}\, e^{\frac{ip_x' x'}{\hbar}}$$

$$= \frac{1}{2\pi\hbar}\int dx\, x\, e^{\frac{ix}{\hbar}(p_x'-p_x)}$$

$$= \frac{i}{2\pi}\int dx\,\frac{\partial}{\partial p_x}e^{\frac{ix}{\hbar}(p_x'-p_x)} = i\hbar\frac{\partial}{\partial p_x}\delta(p_x'-p_x)$$

$$\int dx\, e^{ikx} = 2\pi\delta(k) \rightarrow \int_{-\infty}^{\infty} dx\, e^{\frac{ix}{\hbar}(p'-p)} = 2\pi\,\delta\left(\frac{p'-p}{\hbar}\right) = 2\pi\hbar\,\delta(p'-p)$$

↳ 坐标算符在动量表象 (p_x) 下的表示为 $i\hbar\frac{\partial}{\partial p_x}$

态矢量在 x 和 p 之间的变换（傅里叶变换）

$$\psi(x) = \langle x|\psi\rangle = \langle x|p_x\rangle\langle p_x|\psi\rangle$$

$$= \frac{1}{\sqrt{2\pi\hbar}}\int dp_x\, e^{\frac{ip_x x}{\hbar}}\,\psi(p_x)$$

$$\psi(p_x) = \langle p_x|\psi\rangle = \langle p_x|x\rangle\langle x|\psi\rangle$$

$$= \frac{1}{\sqrt{2\pi\hbar}}\int dx\, e^{\frac{ip_x x}{\hbar}}\,\psi(x)$$

⇒ 总结：

$$\langle\vec r|\vec r'\rangle = \delta(\vec r-\vec r') \qquad \langle\vec p|\vec p'\rangle = \delta(\vec p-\vec p')$$

$$\langle\vec r|\hat{\vec r}|\vec r'\rangle = \vec r\,\delta(\vec r-\vec r') \qquad \langle\vec p|\hat{\vec p}|\vec p'\rangle = \vec p\,\delta(\vec p-\vec p')$$

$$\langle\vec r|\hat{\vec p}|\vec r'\rangle = -i\hbar\nabla\delta(\vec r-\vec r') \qquad \langle\vec p|\hat{\vec r}|\vec p'\rangle = i\hbar\nabla\delta(\vec p-\vec p')$$

$$\langle\vec r|\vec p\rangle = \frac{1}{(2\pi\hbar)^{3/2}}e^{\frac{i(\vec p\cdot\vec r)}{\hbar}} \qquad \langle\vec p|\vec r\rangle = \frac{1}{(2\pi\hbar)^{3/2}}e^{-\frac{i(\vec p\cdot\vec r)}{\hbar}}$$

§ 2.5 运动方程

1. 时间演化算符 $\hat U(t,t)$

$$|\psi(t')\rangle = \hat U(t',t)|\psi(t)\rangle$$

(1) $\hat U$ 只与系统有关，与具体的态无关

$$|\gamma(t)\rangle = c_1|\alpha(t)\rangle + c_2|\beta(t)\rangle$$

$$\hat U|\gamma(t)\rangle = c_1\hat U|\alpha(t)\rangle + c_2\hat U|\beta(t)\rangle$$

态叠加原理 → $\hat U$ 是线性算符.

(2) $\hat U$ 是幺正算符：

归一化条件 $\langle\psi(t)|\hat U^\dagger(t',t)\hat U(t',t)|\psi(t)\rangle$

$$=\langle\psi(t')|\psi(t')\rangle = 1$$

$$\Rightarrow \hat U(t,t')\hat U(t',t) = 1$$

2 schrodinger 运动方程.

$$\hat U(t+\delta t,t) = 1+\delta t\,\Omega(t)$$

$$1 = \hat U^\dagger(t+\delta t,t)\hat U(t+\delta t,t)$$

$$=[1+\delta t\,\Omega^\dagger(t)][1+\delta t\,\Omega(t)]$$

$$\approx 1+\delta t[\Omega(t)+\Omega^\dagger(t)] \quad \rightarrow \quad \Omega(t)\text{ 是反厄米算符.}$$

$$\frac{\partial|\psi(t)\rangle}{\partial t} = \lim_{\delta t\to 0}\frac{|\psi(t+\delta t)\rangle - |\psi(t)\rangle}{\delta t}$$

$$=\lim_{\delta t\to 0}\frac{\hat U(t+\delta t,t)-1}{\delta t} = \Omega(t)$$

$$\hat H(t) = i\hbar\,\Omega(t)$$

$$i\hbar\frac{\partial}{\partial t}|\psi(t)\rangle = \hat H(t)|\psi(t)\rangle$$

在给定表象下：

$$i\hbar\frac{\partial}{\partial t}\psi(\vec{r},t) = \int d^3r' \langle r|\hat{H}|r'\rangle \psi(r',t)$$

坐标表象中势是局域的 $\langle r|\hat{H}|r'\rangle = H(\vec{r},t)\,\delta(\vec{r}-\vec{r}')$? A: $\hat{H}$ 只有"对角元"

没有长程相互作用，非局域~

⇒ Schrodinger 方程： $i\hbar\frac{\partial}{\partial t}\psi(\vec{r},t) = H(\vec{r},t)\,\psi(\vec{r},t)$

3. 时间演化算符的具体形式.

$$\psi(t) = \hat{U}(t,t_0)\,\psi(t_0)$$

$$\Rightarrow \quad i\hbar\frac{\partial}{\partial t}\hat{U}(t,t_0) = \hat{H}\hat{U}(t,t_0)$$

$$\hat{U}(t,t_0) = 1-\frac{i}{\hbar}\int_{t_0}^{t}dt\,\hat{H}\hat{U}(t,t_0)$$

迭代法：

$$\hat{U}(t,t_0) = \sum_{n=1}^{\infty}\left(-\frac{i}{\hbar}\right)^n \int_{t_0}^{t}dt_1\int_{t}^{t_1}dt_2 \;—\; \int_{t}^{t_{n-1}}dt_n\,\hat{H}\hat{U}(t_n,t_0)$$

引入编时算符： 不同时刻 $\hat{H}$ 不一定对易

$$\hat{T}[\hat{H}(t_1)\cdots\hat{H}(t_n)] = \begin{cases}\hat{H}(t_1)\cdots\hat{H}(t_n) & t_1>\cdots>t_n\\ \hat{H}(t_n)\cdots\hat{H}(t_1) & t_n>\cdots>t_1\end{cases}$$

将积分变为：

$$\hat{U}(t,t_0) = \sum_{n=0}^{\infty}\frac{1}{n!}\left(-\frac{i}{\hbar}\right)^n\int_{t_0}^{t}dt_1\int_{t_0}^{t}dt_2 \;—\; \int_{t_0}^{t}dt_n\,\hat{T}[\hat{H}(t_1)\;—\;\hat{H}(t_n)]$$

$$=\hat{T}[e^{-\frac{i}{\hbar}\int_{t_0}^{t}H(t')dt}$$

不含时：

$$\hat{U}(t,t_0) = e^{-\frac{i}{\hbar}\hat{H}(t-t_0)}$$

Hamilton 量本征态，

$$|\phi_n(t)\rangle = e^{-\frac{i}{\hbar}E_n(t-t_0)}|\phi_n(0)\rangle$$

⇒ 研究态矢量随时间的演化，可以在能量表象展开.

$$|\psi(0)\rangle = \sum_n C_n|\phi_n(0)\rangle$$

$$|\psi(t)\rangle = \sum_n C_n|\phi_n(t)\rangle$$

例：考虑 Hilbert 空间和三维系统，基矢为 $\{|1\rangle, |2\rangle, |3\rangle\}$

$$\hat{H} = \hbar\omega\begin{pmatrix}1&0&0\\0&2&0\\0&0&2\end{pmatrix} \quad \hat{A} = a\begin{pmatrix}1&0&0\\0&0&1\\0&1&0\end{pmatrix} \quad \hat{B} = b\begin{pmatrix}0&1&1\\1&1&0\\1&0&1\end{pmatrix}$$

$t=0$ 时刻系统状态为 $|\psi(0)\rangle = \frac{1}{\sqrt{2}}|1\rangle + \frac{1}{2}|2\rangle + \frac{1}{2}|3\rangle$

(1) 计算 $t=0$ 时刻 $\Delta H = \sqrt{\langle\hat{H}^2\rangle - \langle H\rangle^2}$

$$\hat{H}^2 = \hbar^2\omega^2\begin{pmatrix}1&0&0\\0&4&0\\0&0&4\end{pmatrix}$$

$$\langle\hat{H}^2\rangle = \langle\psi|\hat{H}^2|\psi\rangle = \hbar^2\omega^2\left(\frac{1}{\sqrt{2}}\ \frac{1}{2}\ \frac{1}{2}\right)\begin{pmatrix}1&0&0\\0&4&0\\0&0&4\end{pmatrix}\begin{pmatrix}\frac{1}{\sqrt{2}}\\\frac{1}{2}\\\frac{1}{2}\end{pmatrix}$$

$$= \hbar^2\omega^2\left(\frac{1}{\sqrt{2}}\ 2\ 2\right)\begin{pmatrix}\frac{1}{\sqrt{2}}\\\frac{1}{2}\\\frac{1}{2}\end{pmatrix} = \frac{5}{2}\hbar^2\omega^2$$

$$\langle\hat{H}\rangle = \langle\psi|\hat{H}|\psi\rangle = \hbar\omega\left(\frac{1}{\sqrt{2}}\ \frac{1}{2}\ \frac{1}{2}\right)\begin{pmatrix}1&0&0\\0&2&0\\0&0&2\end{pmatrix}\begin{pmatrix}\frac{1}{\sqrt{2}}\\\frac{1}{2}\\\frac{1}{2}\end{pmatrix}$$

$$= \hbar\omega\left(\frac{1}{\sqrt{2}}\ 1\ 1\right)\begin{pmatrix}\frac{1}{\sqrt{2}}\\\frac{1}{2}\\\frac{1}{2}\end{pmatrix} = \frac{3}{2}\hbar\omega$$

$$\Delta H = \sqrt{\langle\hat{H}^2\rangle - \langle\hat{H}\rangle^2} = \frac{1}{2}\hbar\omega$$

(2) $$|\hat{A} - \lambda a I| = a^3\begin{vmatrix}1-\lambda&0&0\\0&-\lambda&1\\0&1&-\lambda\end{vmatrix} = a^3[(1-\lambda)(\lambda^2-1)] = 0$$

$\Rightarrow \lambda = 1$ 或 $\lambda = -1$

$\lambda = 1$ 时，$\begin{pmatrix}0&0&0\\0&-1&1\\0&1&-1\end{pmatrix} \to \begin{pmatrix}0&-1&1\\0&0&0\\0&0&0\end{pmatrix}$ $|\alpha_1\rangle = \begin{pmatrix}1\\0\\0\end{pmatrix}$ $|\alpha_2\rangle = \frac{1}{\sqrt{2}}\begin{pmatrix}0\\1\\1\end{pmatrix}$

$\lambda = -1$ 时，$\begin{pmatrix}2&0&0\\0&1&1\\0&1&1\end{pmatrix} \to \begin{pmatrix}2&0&0\\0&1&1\\0&0&0\end{pmatrix}$ $|\alpha_3\rangle = \frac{1}{\sqrt{2}}\begin{pmatrix}0\\1\\-1\end{pmatrix}$

$$a_1 = \langle\alpha_1|\psi(0)\rangle^2 + \langle\alpha_2|\psi(0)\rangle^2 = \frac{1}{2} + \frac{1}{2}\left(\frac{1}{2}+\frac{1}{2}\right) = 1$$

$$a_2 = \langle\alpha_3|\psi(0)\rangle^2 = 0$$

(3) $$\hat{A}\hat{B} = ab\begin{pmatrix}1&0&0\\0&0&1\\0&1&0\end{pmatrix}\begin{pmatrix}0&1&1\\1&1&0\\1&0&1\end{pmatrix} = ab\begin{pmatrix}0&1&1\\1&0&1\\1&1&0\end{pmatrix}$$

$$\hat{B}\hat{A} = ba\begin{pmatrix}0&1&1\\1&1&0\\1&0&1\end{pmatrix}\begin{pmatrix}1&0&0\\0&0&1\\0&1&0\end{pmatrix} = ab\begin{pmatrix}0&1&1\\1&0&1\\1&1&0\end{pmatrix}$$

$$|B-\lambda b I| = b\begin{vmatrix} -\lambda & 1 & 1 \\ 1 & 1-\lambda & 0 \\ 1 & 0 & 1-\lambda \end{vmatrix} = -\lambda(1-\lambda)^2-(1-\lambda)-(1-\lambda) = [-\lambda(1-\lambda)-2](1-\lambda)$$

$$= (\lambda^2-\lambda-2)(1-\lambda) = (\lambda+1)(\lambda-2)(1-\lambda)$$

$$\lambda=-1 \quad \begin{pmatrix} 1 & 1 & 1 \\ 1 & 2 & 0 \\ 1 & 0 & 2 \end{pmatrix} \to \begin{pmatrix} 0 & 0 & 0 \\ 1 & 2 & 0 \\ 1 & 0 & 2 \end{pmatrix} \to \begin{pmatrix} 1 & 2 & 0 \\ 0 & -2 & 2 \\ 0 & 0 & 0 \end{pmatrix} \quad |\beta_1\rangle = \frac{1}{\sqrt{6}}\begin{pmatrix} 2 \\ -1 \\ -1 \end{pmatrix}$$

$$\lambda=2 \quad \begin{pmatrix} -2 & 1 & 1 \\ 1 & -1 & 0 \\ 1 & 0 & -1 \end{pmatrix} \to \begin{pmatrix} 1 & -1 & 0 \\ 1 & 0 & -1 \\ 0 & 0 & 0 \end{pmatrix} \to \begin{pmatrix} 1 & -1 & 0 \\ 0 & 1 & -1 \\ 0 & 0 & 0 \end{pmatrix} \quad |\beta_2\rangle = \frac{1}{\sqrt{3}}\begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}$$

$$\lambda=1 \quad \begin{pmatrix} -1 & 1 & 1 \\ 1 & 0 & 0 \\ 1 & 0 & 0 \end{pmatrix} \to \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{pmatrix} \quad |\beta_3\rangle = \frac{1}{\sqrt{2}}\begin{pmatrix} 0 \\ 1 \\ -1 \end{pmatrix}$$

共同本征态 $|\beta_3\rangle$ 与 $|\alpha_3\rangle$，继续求出剩下两个

在 A 的本征子空间中：

① $\lambda = 1$

$$|\alpha_1\rangle = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} \quad |\alpha_2\rangle = \frac{1}{\sqrt{2}}\begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix}$$

B 的限制：$B' = \begin{pmatrix} \langle\alpha_1|\hat{B}|\alpha_1\rangle & \langle\alpha_1|\hat{B}|\alpha_2\rangle \\ \langle\alpha_2|\hat{B}|\alpha_1\rangle & \langle\alpha_2|\hat{B}|\alpha_2\rangle \end{pmatrix} = \begin{pmatrix} 0 & \sqrt{2} \\ \sqrt{2} & 1 \end{pmatrix}$

$$|B'-\lambda I| = \begin{vmatrix} -\lambda & \sqrt{2} \\ \sqrt{2} & 1-\lambda \end{vmatrix} = \lambda(\lambda-1)-2 = (\lambda-2)(\lambda+1)$$

$$\lambda_1 = 2 \quad \begin{pmatrix} -2 & \sqrt{2} \\ \sqrt{2} & -1 \end{pmatrix} \to |\beta_1'\rangle = \frac{1}{\sqrt{3}}\begin{pmatrix} 1 \\ \sqrt{2} \end{pmatrix}$$

$$\lambda_2 = -1 \quad \begin{pmatrix} 1 & \sqrt{2} \\ \sqrt{2} & 2 \end{pmatrix} \to |\beta_2'\rangle = \frac{1}{\sqrt{3}}\begin{pmatrix} \sqrt{2} \\ -1 \end{pmatrix}$$

$\Rightarrow$ A、B 的共同本征态：$\sqrt{\frac{1}{3}}|\alpha_1\rangle + \sqrt{\frac{2}{3}}|\alpha_2\rangle$ 和 $\sqrt{\frac{2}{3}}|\alpha_1\rangle - \sqrt{\frac{1}{3}}|\alpha_2\rangle$

② $\lambda = -1 \quad |\alpha_3\rangle = \frac{1}{\sqrt{2}}\begin{pmatrix} 0 \\ 1 \\ -1 \end{pmatrix}$

$$B'' = \langle\alpha_3|\hat{B}|\alpha_3\rangle = \frac{1}{2}(0\ 1\ -1)\begin{pmatrix} 0 & 1 & 1 \\ 1 & 1 & 0 \\ 1 & 0 & 1 \end{pmatrix}\begin{pmatrix} 0 \\ 1 \\ -1 \end{pmatrix} = \frac{1}{2}(0\ 1\ -1)\begin{pmatrix} 0 \\ 1 \\ -1 \end{pmatrix} = 1$$

$\Rightarrow$ A、B 的共同本征态：$|\alpha_3\rangle$

(4) $|\gamma_1\rangle = \sqrt{\frac{1}{3}}|\alpha_1\rangle + \sqrt{\frac{2}{3}}|\alpha_2\rangle = \frac{\sqrt{3}}{3}\begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}$

$$|\gamma_2\rangle = \sqrt{\frac{2}{3}}|\alpha_1\rangle - \sqrt{\frac{1}{3}}|\alpha_2\rangle = \frac{\sqrt{3}}{3}\begin{pmatrix} \sqrt{2} \\ -\frac{\sqrt{2}}{2} \\ -\frac{\sqrt{2}}{2} \end{pmatrix} = \frac{\sqrt{6}}{6}\begin{pmatrix} 2 \\ -1 \\ -1 \end{pmatrix}$$

$$|\gamma_3\rangle = |\alpha_3\rangle = \frac{1}{\sqrt{2}}\begin{pmatrix} 0 \\ 1 \\ -1 \end{pmatrix}$$

$$U = (|\gamma_1\rangle, |\gamma_2\rangle, |\gamma_3\rangle) = \begin{pmatrix} \frac{\sqrt{3}}{3} & \frac{\sqrt{6}}{3} & 0 \\ \frac{\sqrt{3}}{3} & -\frac{\sqrt{6}}{6} & \frac{\sqrt{2}}{2} \\ \frac{\sqrt{3}}{3} & -\frac{\sqrt{6}}{6} & -\frac{\sqrt{2}}{2} \end{pmatrix}$$

$$\vec{H}' = U^{+}\vec{H}U$$

$$= \begin{pmatrix} \frac{\sqrt{3}}{3} & \frac{\sqrt{3}}{3} & \frac{\sqrt{3}}{3} \\ \frac{\sqrt{6}}{3} & -\frac{\sqrt{6}}{6} & -\frac{\sqrt{6}}{6} \\ 0 & \sqrt{2} & \sqrt{2} \end{pmatrix} \hbar\omega \begin{pmatrix} 1 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 2 \end{pmatrix} \begin{pmatrix} \frac{\sqrt{3}}{3} & \frac{\sqrt{6}}{3} & 0 \\ \frac{\sqrt{3}}{3} & -\frac{\sqrt{6}}{6} & \frac{\sqrt{2}}{2} \\ \sqrt{3} & \sqrt{6} & -\sqrt{2} \end{pmatrix}$$

$$= \hbar\omega \begin{pmatrix} \frac{\sqrt{3}}{3} & \frac{2\sqrt{3}}{3} & \frac{2\sqrt{3}}{3} \\ \frac{\sqrt{6}}{3} & -\frac{\sqrt{6}}{3} & -\frac{\sqrt{6}}{3} \\ 0 & \sqrt{2} & -\sqrt{2} \end{pmatrix} \begin{pmatrix} \frac{\sqrt{3}}{3} & \frac{\sqrt{6}}{3} & 0 \\ \frac{\sqrt{3}}{2} & -\frac{\sqrt{6}}{6} & \frac{\sqrt{2}}{2} \\ \frac{\sqrt{3}}{3} & -\frac{\sqrt{6}}{6} & -\frac{\sqrt{2}}{2} \end{pmatrix}$$

$$= \hbar\omega \begin{pmatrix} \frac{5}{3} & -\frac{\sqrt{2}}{3} & 0 \\ -\frac{\sqrt{2}}{3} & -2 & 0 \\ 0 & 0 & 2 \end{pmatrix}$$

(5) $|\alpha_1\rangle = \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix}$ $|\alpha_2\rangle = \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix}$ $|\alpha_3\rangle = \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}$

$$|\psi(0)\rangle = \langle\alpha_1|\psi(0)\rangle|\alpha_1\rangle + \cdots$$

$$= 1\,|\alpha_1\rangle + \left(\frac{1}{\sqrt{2}} + \frac{1}{2}\right)|\alpha_2\rangle + \left(\frac{1}{\sqrt{2}} + \frac{1}{2}\right)|\alpha_3\rangle$$

$$|\psi(t)\rangle = \frac{1}{\sqrt{2}}|1\rangle e^{-i\omega t} + \frac{1}{2}|2\rangle e^{-i2\omega t} + \frac{1}{2}|3\rangle e^{-i2\omega t}$$

$$\langle\vec{B}(t)\rangle = \langle\psi(t)|\vec{B}(t)|\psi(t)\rangle$$

$$= \left(\frac{1}{\sqrt{2}}e^{i\omega t} \quad \frac{1}{2}e^{i2\omega t} \quad \frac{1}{2}e^{i2\omega t}\right) b \begin{pmatrix} 0 & 1 & 1 \\ 1 & 1 & 0 \\ 1 & 0 & 1 \end{pmatrix} \begin{pmatrix} \frac{1}{\sqrt{2}}e^{-i\omega t} \\ \frac{1}{2}e^{-i2\omega t} \\ \frac{1}{2}e^{-i2\omega t} \end{pmatrix}$$

$$= b\left(e^{i2\omega t} \quad \frac{1}{\sqrt{2}}e^{i\omega t} + \frac{1}{2}e^{i2\omega t} \quad \frac{1}{\sqrt{2}}e^{i\omega t} + \frac{1}{2}e^{i2\omega t}\right) \begin{pmatrix} \frac{1}{\sqrt{2}}e^{-i\omega t} \\ \frac{1}{2}e^{-i2\omega t} \\ \frac{1}{2}e^{-i2\omega t} \end{pmatrix}$$

$$= b\left[\frac{1}{\sqrt{2}}e^{i\omega t} + 2\left(\frac{1}{2\sqrt{2}}e^{-i\omega t} + \frac{1}{4}\right)\right]$$

$$= b\left(\frac{1}{2} + \sqrt{2}\cos\omega t\right)$$

作业.

1. $\hat{\vec{L}} = \hat{L}_x \vec{e_x} + \hat{L}_y \vec{e_y} + \hat{L}_z \vec{e_z}$, $\hat{\vec{r}} = \hat{x}\vec{e_x} + \hat{y}\vec{e_y} + \hat{z}\vec{e_z}$

(1) $[\hat{x}, \hat{L}_x] = [\hat{x}, \hat{y}\hat{p}_z - \hat{z}\hat{p}_y] = [\hat{x}, \hat{y}\hat{p}_z] - [\hat{x}, \hat{z}\hat{p}_y] = 0$

$[\hat{x}, \hat{L}_y] = [\hat{x}, \hat{z}\hat{p}_x - \hat{x}\hat{p}_z] = [\hat{x}, \hat{z}\hat{p}_x] - [\hat{x}, \hat{x}\hat{p}_z] = -i\hbar\hat{z}$

$[\hat{x}, \hat{L}_z] = [\hat{x}, \hat{x}\hat{p}_y - \hat{y}\hat{p}_x] = -i\hbar\hat{y}$

(2) 证明：$[\hat{\vec{r}}, \hat{L}^2] = \alpha\hat{\vec{r}} \times \hat{L} + \beta\hat{\vec{r}}$ ，给出 α, β 的值.

$[\hat{\vec{r}}, \hat{L}^2] = [r_i, L_j L_j]$

$= 2L_j[r_i, L_j] = L_j[r_i, \varepsilon_{jkm} r_k p_m] + [r_i, \varepsilon_{jkm} r_k p_m] L_j$

$= i\hbar\,\delta_{im}\varepsilon_{jkm}(r_k L_j + L_j r_k) = i\hbar\,\varepsilon_{ijk}(r_k L_j + L_j r_k)$

$= 2i\hbar\,\varepsilon_{ijk}(r_k L_j) + i\hbar\,\varepsilon_{ijk}(L_j r_k - r_k L_j)$

$= -2i\hbar(\hat{\vec{r}} \times \hat{L})_i - i\hbar\,\varepsilon_{ijk}[r_k, L_j] = -2i\hbar(\hat{\vec{r}} \times \hat{L})_i - i\hbar\,\varepsilon_{ijk}[r_k, \varepsilon_{jml} r_m p_l]$

$= -2i\hbar(\hat{\vec{r}} \times \hat{L})_i + \hbar^2 r_m \delta_{kl}\varepsilon_{ijk}\varepsilon_{jml}$

$= -2i\hbar(\hat{\vec{r}} \times \hat{L})_i - \hbar^2 \varepsilon_{ijk}\varepsilon_{jkm} r_m$

$= -2i\hbar(\hat{\vec{r}} \times \hat{L})_i - \hbar^2(\delta_{im}\delta_{jj} - \delta_{ij}\delta_{mj}) r_m$

$= -2i\hbar(\hat{\vec{r}} \times \hat{L})_i - i\hbar(3-1)\delta_{im} r_i$

$= -2i\hbar(\hat{\vec{r}} \times \hat{L})_i - 2\hbar^2 r_i$

2. (1) $\hat{A}|\psi\rangle = \lambda|\psi\rangle$, $\hat{A}^2|\psi\rangle = \lambda^2|\psi\rangle = |\psi\rangle$

$\lambda = \pm 1$ → 二维 Hilbert 空间.

在A表象：$\hat{A} = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$ 若本征值不简并，必须一个1，一个-1

$\hat{B} = (a \quad b)$

$c \quad d$

$$\begin{cases} \vec{B}^{+} = B \\ \vec{B}^{2} = 1 \\ \{\vec{A}, \vec{B}\} = \vec{A}\vec{B} + \vec{B}\vec{A} = 0 \end{cases}$$

$$\begin{cases} b^{*} = c \\ a^2 + bc = d^2 + bc = 1 , \quad b(a+d) = c(a+d) = 0 \\ \begin{pmatrix} 2a & 0 \\ 0 & 2d \end{pmatrix} = 0 \rightarrow a = d = 0 \end{cases} \Rightarrow \vec{B} = \begin{pmatrix} 0 & e^{i\varphi} \\ e^{-i\varphi} & 0 \end{pmatrix}$$

(2) 在A表象中，$|\lambda_{A1}\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$，$|\lambda_{A2}\rangle = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$

$$|B - \lambda I| = \begin{vmatrix} -\lambda & e^{i\varphi} \\ e^{-i\varphi} & -\lambda \end{vmatrix} = (\lambda^2 - 1) = 0$$

$$\lambda = 1 , \quad \begin{pmatrix} -1 & e^{i\varphi} \\ e^{-i\varphi} & -1 \end{pmatrix} \rightarrow \begin{pmatrix} -1 & e^{i\varphi} \\ 0 & 0 \end{pmatrix} \quad |\lambda_{B1}\rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ e^{-i\varphi} \end{pmatrix}$$

$$\lambda = -1 , \quad \begin{pmatrix} 1 & e^{i\varphi} \\ e^{-i\varphi} & 1 \end{pmatrix} \rightarrow \begin{pmatrix} 1 & e^{i\varphi} \\ 0 & 0 \end{pmatrix} \quad |\lambda_{B2}\rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ -e^{-i\varphi} \end{pmatrix}$$

变换矩阵 $U_{ij} = \langle \lambda_{Bi} | \lambda_{Aj} \rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & e^{-i\varphi} \\ 1 & -e^{-i\varphi} \end{pmatrix}$

$$A' = U A U^{+}$$

$$= \frac{1}{2} \begin{pmatrix} 1 & e^{-i\varphi} \\ 1 & -e^{-i\varphi} \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \begin{pmatrix} 1 & 1 \\ e^{i\varphi} & -e^{i\varphi} \end{pmatrix}$$

$$= \frac{1}{2} \begin{pmatrix} 1 & e^{-i\varphi} \\ 1 & -e^{-i\varphi} \end{pmatrix} \begin{pmatrix} 1 & 1 \\ -e^{i\varphi} & e^{i\varphi} \end{pmatrix}$$

$$= \frac{1}{2} \begin{pmatrix} 1-1 & 2 \\ 2 & 1-1 \end{pmatrix} = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$$

(3) $C = \begin{pmatrix} a' & b' \\ c' & d' \end{pmatrix}$

$$\begin{cases} \hat{C}^2 = 1 \\ [\hat{B}, \hat{C}] = \hat{B}\hat{C} - \hat{C}\hat{B} = 0 \end{cases}$$

$$\begin{cases} a'^2 + b'c' = d'^2 + b'c' = 1 , \quad b'(a'+d') = c'(a'+d') \\ a' = d' , \quad \frac{b'}{c'} = e^{2i\varphi} \end{cases}$$

$$\begin{pmatrix} 0 & e^{i\varphi} \\ e^{-i\varphi} & 0 \end{pmatrix} \begin{pmatrix} a' & b' \\ c' & d' \end{pmatrix} - \begin{pmatrix} a' & b' \\ c' & d' \end{pmatrix} \begin{pmatrix} 0 & e^{i\varphi} \\ e^{-i\varphi} & 0 \end{pmatrix}$$

$$= \begin{pmatrix} c'e^{i\varphi} - b'e^{-i\varphi} & d'e^{i\varphi} - a'e^{i\varphi} \\ a'e^{-i\varphi} - d'e^{-i\varphi} & b'e^{-i\varphi} - c'e^{i\varphi} \end{pmatrix} = 0$$

$$\hat{C} = \begin{pmatrix} 0 & -e^{-\varphi} \\ -e^{-i\varphi} & 0 \end{pmatrix}$$

3.

$$\langle x|\hat{A}|x'\rangle = x^2 \frac{\partial}{\partial x}\delta(x-x')$$

$$\langle x|\hat{A}^{\dagger}|x'\rangle = [\langle x'|\hat{A}|x\rangle]^{\dagger} = x'^2 \frac{\partial}{\partial x'}\delta(x'-x)$$

$$= -x'^2 \frac{\partial}{\partial x}\delta(x'-x) = -x'^2 \frac{\partial}{\partial x}\delta(x-x')$$

$$\langle p|\hat{A}^{\dagger}|p'\rangle = \langle p|x\rangle\langle x|\hat{A}^{\dagger}|x'\rangle\langle x'|p'\rangle$$

$$= \frac{1}{2\pi\hbar}\iint dx\,dx'\; e^{-\frac{ipx}{\hbar}}\left[-x'^2\frac{\partial}{\partial x}\delta(x-x')\right]e^{\frac{ip'x'}{\hbar}}$$

$$= \frac{1}{2\pi\hbar}\int dx\; e^{-\frac{ipx}{\hbar}}\, x^2 \frac{\partial}{\partial x'} e^{\frac{ip'x'}{\hbar}}$$

$$= \frac{1}{2\pi\hbar}\frac{ip'}{\hbar}\int dx\; x^2\, e^{\frac{ix(p-p')}{\hbar}}$$

$$= \frac{ip'}{2\pi\hbar^2}\int dx\left(\frac{\hbar}{i}\right)^2\left(\frac{\partial}{\partial p}\right)^2 e^{\frac{ix(p-p')}{\hbar}}$$

$\int dx\, e^{ikx} = 2\pi\delta(k)$

$$= -\frac{ip'}{2\pi}\left(\frac{\partial}{\partial p}\right)^2\cdot 2\pi\,\delta\left(\frac{p-p'}{\hbar}\right)$$

$$= -ip'\hbar\left(\frac{\partial}{\partial p}\right)^2\delta(p-p')$$

$$* \quad \langle x|\hat{p}|x'\rangle = -i\hbar\frac{\partial}{\partial x}\delta(x-x')$$

$$\hat{A} = \frac{\hat{x}^2}{-i\hbar}\hat{p} = \frac{i}{\hbar}\hat{x}^2\hat{p}$$

$$\hat{A}^{\dagger} = -\frac{i}{\hbar}(\hat{x}^2\hat{p})^{\dagger} = -\frac{i}{\hbar}\hat{p}\hat{x}^2$$

$$\langle x|\hat{A}|x'\rangle = \frac{\partial}{\partial x}\left[x^2\delta(x-x')\right]$$

$$= 2x\,\delta(x-x') + x^2\frac{\partial}{\partial x}\delta(x-x')$$

$$= 2x'\delta(x'-x) - x'^2\frac{\partial}{\partial x'}\delta(x-x')$$

$$= 2x'\,\delta(x'-x) - x'^2\frac{\partial}{\partial x'}\delta(x'-x)$$

4. $[\hat{p}, f(\hat{x})]$

$= [\hat{p}, \sum_i c_i \hat{x}^i] = \sum_i c_i [\hat{p}, \hat{x}^i]$

$= \sum_i c_i (\hat{x} [\hat{p}, \hat{x}^{i-1}] - i\hbar \hat{x}^{i-1})$

$= \sum_i c_i (-n i\hbar \hat{x}^{i-1}) = -i\hbar \frac{\partial f(\hat{x})}{\partial \hat{x}}$

$[\hat{x}, f(\hat{p})]$

$= [\hat{x}, \sum_i c_i' \hat{p}^i] = \sum c_i [\hat{x}, \hat{p}^i]$

$= i\hbar \frac{\partial f(\hat{p})}{\partial \hat{p}}$

$[\hat{p}, \hat{x}^i] = \hat{x} [\hat{p}, \hat{x}^{i-1}] + \hat{x}^{i-1} [\hat{p}, \hat{x}]$

$= \hat{x} [\hat{p}, \hat{x}^{i-1}] - i\hbar \hat{x}^{i-1}$